

E219288030_1

The equation $x^4 + xy + x + 2y^3 + y + 1 = 0$ has no integer solutions

Size $h = 41$, length $l = 12$

Theorem 1. *The equation*

$$x^4 + xy + x + 2y^3 + y + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses three polynomial identities of the form $H_k T_k = U_k^2 - 2V_k^2$ on the curve (1), and the fact that an odd prime p with $(2/p) = -1$ divides $U_k^2 - 2V_k^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm form $a^2 - 2b^2$

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right) \left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(2/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 - 2b^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv 2(bc)^2 \equiv 2 \pmod{p}$, so 2 is a square modulo p . As $(2/p) \neq 0$, it is a non-zero square, so $(2/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 - 2b^2 \neq 0$, and $v_p(a^2 - 2b^2)$ is even for every odd prime p with $(2/p) = -1$.*

Proof. Since 2 is not the square of an integer, $a^2 - 2b^2 \neq 0$. Let p be an odd prime with $(2/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 - 2b'^2$ by Lemma 2. Since $a^2 - 2b^2 = p^{2k}(a'^2 - 2b'^2)$, we get $v_p(a^2 - 2b^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{8}$, or with $n' \equiv -n \pmod{8}$. Then $(2/n') = (2/n)$.*

Proof. Write $2 = \varepsilon 2^e q$ with $\varepsilon = 1$, $e = 1$ and $q = 1$. By multiplicativity and the reciprocity laws,

$$\left(\frac{2}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since 8 is divisible by $4q$, and by $8q$, the first claim follows. If $n' \equiv -n \pmod{8}$, then, as $\varepsilon = 1$, the factor (ε/n) does not change, $(2/n)$ does not change because $n'^2 \equiv n^2 \pmod{16}$, and each of $(-1)^{(n-1)(q-1)/4}$ and (n/q) is multiplied by $(-1)^{(q-1)/2}$, because $(n' - 1)/2 \equiv (n - 1)/2 + 1 \pmod{2}$ and $(n'/q) = (-1/q)(n/q) = (-1)^{(q-1)/2}(n/q)$. This proves the second claim. $\square$

Lemma 5. *Let n be a non-zero integer, and let s , u and m be positive integers such that every prime factor of s divides 4, both $2s$ and $8s$ divide m , u is odd, $(2/u) = -1$, and $n \equiv su \pmod{m}$. Then there is an odd prime p with $(2/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by 8, n' is odd and $n' \equiv u \pmod{8}$. Since $|n'| \equiv \pm u \pmod{8}$, Lemma 4 gives $(2/|n'|) = (2/u) = -1$. By multiplicativity, the product of $(2/p)^{v_p(n')}$ over the prime factors p of $|n'|$ is -1 , so some prime factor p of n' satisfies $(2/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 4$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + xy + x + 2y^3 + y + 1$, so that (1) is the equation $F(x, y) = 0$. For the first identity, we define

$$\begin{aligned} H_1 &= -y - 1, \\ T_1 &= x + 2y^2 - 2y + 3, \\ U_1 &= x^2, \\ V_1 &= 1, \\ G_1 &= -1, \\ A_1 &= 0, \\ B_1 &= 1, \end{aligned}$$

for the second identity, we define

$$\begin{aligned} H_2 &= x - y, \\ T_2 &= 2x^2 + 2xy + x + 2y^2 + 1, \\ U_2 &= x^2 + x + 1, \\ V_2 &= x, \\ G_2 &= -1, \\ A_2 &= 1, \\ B_2 &= -x - 1, \end{aligned}$$

for the third identity, we define

$$\begin{aligned} H_3 &= 2x - y + 1, \\ T_3 &= 8x^2 + 4xy + 9x + 2y^2 + 2y + 3, \end{aligned}$$

$$\begin{aligned}
U_3 &= x^2 + 8x + 6, \\
V_3 &= 5x + 4, \\
G_3 &= -1, \\
A_3 &= 25, \\
B_3 &= -5x - 36.
\end{aligned}$$

Expanding both sides, one checks the identities

$$H_1 T_1 - U_1^2 + 2V_1^2 = G_1 F, \quad (2)$$

$$H_2 T_2 - U_2^2 + 2V_2^2 = G_2 F, \quad (3)$$

$$H_3 T_3 - U_3^2 + 2V_3^2 = G_3 F, \quad (4)$$

$$A_1 U_1 + B_1 V_1 = 1, \quad (5)$$

$$A_2 U_2 + B_2 V_2 = 1, \quad (6)$$

$$A_3 U_3 + B_3 V_3 = 2 \cdot 3. \quad (7)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2), (3), and (4) give

$$H_k T_k = U_k^2 - 2V_k^2 \quad (k = 1, 2, 3), \quad (8)$$

where the polynomials now denote their values at (x, y) .

Step 1: congruences. Checking the 32^2 residue pairs modulo 32, we find that (1) has exactly 16 solutions modulo 32, and that at each of them $H_1 \equiv 6, 10, 22$ or $26 \pmod{32}$, $H_2 \equiv 5, 13, 21$ or $29 \pmod{32}$, or $H_3 \equiv 12 \pmod{32}$. For $k = 1$, we write each residue as su with s and u positive and u odd: $6 = 2 \cdot 3$ with $(2/3) = -1$, $10 = 2 \cdot 5$ with $(2/5) = -1$, $22 = 2 \cdot 11$ with $(2/11) = -1$, and $26 = 2 \cdot 13$ with $(2/13) = -1$. Here for $s = 2$, every prime factor of s divides 4, and $2s = 4$ and $8s = 16$ divide 32. For $k = 2$, these residues are odd, and $(2/5) = -1$, $(2/13) = -1$, $(2/21) = -1$, and $(2/29) = -1$. For $k = 3$, we write this residue as su with s and u positive and u odd: $12 = 4 \cdot 3$ with $(2/3) = -1$. Here for $s = 4$, every prime factor of s divides 4, and $2s = 8$ and $8s = 32$ divide 32.

Step 2: signs. *The value of H_1 .* By (5), U_1 and V_1 are not both zero, so $U_1^2 - 2V_1^2 \neq 0$ by Lemma 3, and (8) gives $H_1 \neq 0$.

The value of H_2 . By (6), U_2 and V_2 are not both zero, so $U_2^2 - 2V_2^2 \neq 0$ by Lemma 3, and (8) gives $H_2 \neq 0$.

The value of H_3 . By (7), U_3 and V_3 are not both zero, so $U_3^2 - 2V_3^2 \neq 0$ by Lemma 3, and (8) gives $H_3 \neq 0$.

Step 3: primes p with $(2/p) = -1$. Let $k \in \{1, 2, 3\}$, and let p be an odd prime with $(2/p) = -1$. We show that $v_p(H_k)$ is even. By (5)–(7), U_k and V_k are not both zero, so (8) and Lemma 3 show that $v_p(H_k T_k) = v_p(U_k^2 - 2V_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 - 2V_k^2$ by (8), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_k U_k + B_k V_k$. For $k = 1$, we have $R_1 = 1$. This is impossible. For $k = 2$, we have $R_2 = 1$. This is impossible. For $k = 3$, we have $R_3 = 2 \cdot 3$. As p is odd, we get $p = 3$. Checking the 3^2 residue pairs modulo 3, we find that F , H_3 and T_3 have no common zero modulo 3, a contradiction. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_k T_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Step 1, there is k such that $H_k \equiv su \pmod{32}$ with s and u as in Lemma 5, and $H_k \neq 0$ by Step 2. Hence Lemma 5 applies to $n = H_k$, so there is an odd prime p with $(2/p) = -1$ such that $v_p(H_k)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.